HEIQ XMEMBRANE PROJECT TEAM

Final result and decision report

EO integrated only in the ERV membrane/core

31 August 2026 · Version 9

Controlled final project basis

This report supersedes Version 8 and all earlier architecture comparisons for current decisions. The proposed EO function is located **inside the ERV membrane/core**; no downstream or post-ERV EO module is part of the selected architecture. Older calculations are retained only as provenance.

Final model-phase conclusion. The matched model identifies a conditional incremental energy window for EO integrated in the ERV. O30 is reachable below the 9,000-m^2 area gate at 5,899 m^2, with 418.400 kW known power and a 125.340-kW (23.05%) window versus the same passive ERV+AC system with EO off. This is strong enough to justify material and coupon testing, but not a validated net-system saving.

Fixed service	O80 inlet at 32 °C/80% RH to 20 °C/50% RH supply
Primary comparator	Passive ERV+AC = 543.741 kW; AC only = 1,166.400 kW
Selected integrated point	O30, 1.00 V, 5,899 m^2, 418.400 kW known subtotal
Calculated EO break-even	51.955 Wh/kg all-in active-core energy
Release status	Controlled V9; source, selected results, manifest and SHA256 checksums packaged
Claim status	Conditional model evidence; no validated membrane, module or commercial saving

Executive summary

The largest established model reduction comes from the passive ERV: the equal-service subtotal falls from 1,166.400 kW for AC alone to 543.741 kW for passive ERV+AC. EO must therefore be judged incrementally against the same ERV with EO off. Crediting the entire AC-to-active-system difference to EO would overstate its contribution.

Passive ERV already reaches 13.215 g/kg, below the O50 coordinate of 14.952 g/kg. O50 therefore has zero EO duty in the matched model. At O30 (8.886 g/kg), the selected area-constrained integrated ERV+EO point uses 5,899 m^2. It combines 316.910 kW of remaining AC with 101.490 kW of known active-core power, for 418.400 kW total. The incremental window is 125.340 kW versus passive ERV+AC and 747.999 kW versus AC alone; only the former is the EO-specific comparison.

The O30 break-even is calculated, not assumed:

$$e_{\mathrm{BE}} = \frac{P_{\text{passive ERV+AC}} - P_{\text{remaining AC}}}{\dot{m}_{\text{water,integrated core}}} = 51.955\ \mathrm{Wh/kg}.$$

The known active-core SEC is 23.246 Wh/kg, leaving 28.709 Wh/kg or 125.340 kW for omitted receiver, heat-rejection and hardware loads before the incremental advantage disappears. The historical 180-Wh/kg shorthand is not the architecture-specific break-even.

Full humidity removal is not available below the current area gate. The minimum valid screened full-duty point requires approximately 17,253 m^2 and 113.7 kW known total power. The lowest-power stored full-duty mechanism coordinate uses 34,425 m^2 and 107.856 kW. Both are outside the accepted area envelope. The practical opportunity is therefore partial active enhancement of the ERV, with O30 the best current under-gate coordinate, rather than forcing EO to complete all dehumidification.

5,899 m^2
Selected O30 area
Inside the 9,000-m^2 gate

125.3 kW
Incremental window
Versus the same ERV with EO off

52.0 Wh/kg
Calculated break-even
Known active core: 23.2 Wh/kg

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1 Architecture and comparison discipline

ONE INLET, ONE ENDPOINT, ONE MATCHED PASSIVE CONTROL

The active and passive cases share the same outdoor inlet, supply endpoint, airflow and sensible-effectiveness basis. EO acts only across the active membrane inside the ERV core. The downstream coil removes the remaining sensible and latent load; at the full humidity endpoint it is sensible only. O50 and O30 are intermediate humidity-ratio coordinates, not alternative weather states and not supply-air definitions.

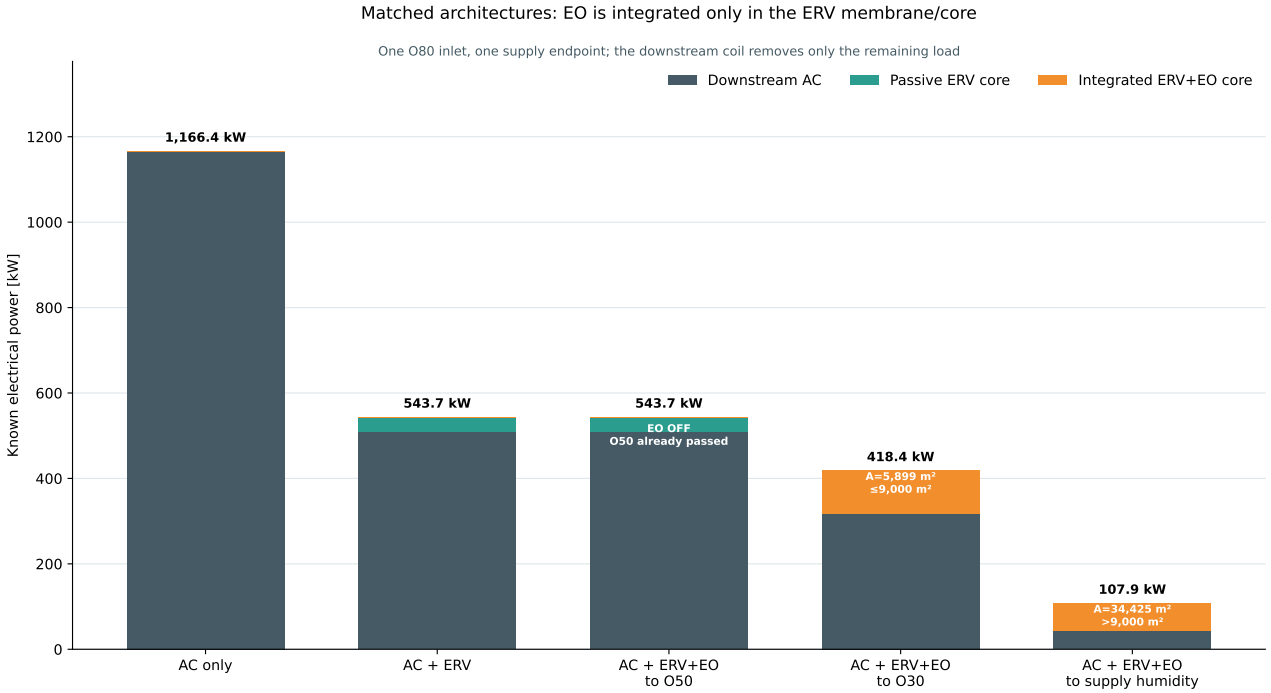


Figure 1: Matched equal-service architectures. EO exists only in the ERV membrane/core. The O30 selection is below 9,000 m^2; the full-duty point is an unconstrained mechanism coordinate.

Comparator rule. Use AC+ERV versus AC to quantify passive recovery. Use AC+ERV+EO versus the same AC+ERV case to quantify incremental EO benefit. The AC-only comparison may be reported separately, but must not be labelled as an EO-only saving.

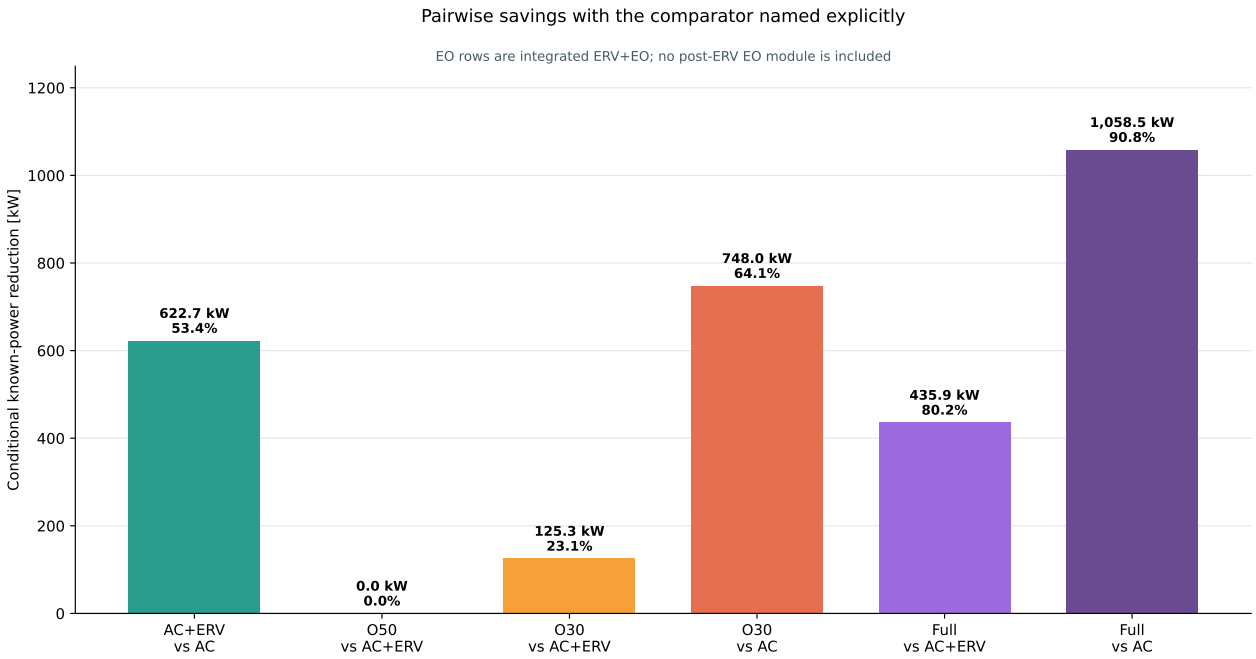


Figure 2: Pairwise conditional reductions with the comparator named explicitly.

2 Results below the 9,000-m^2 gate

FIVE DISTINCT ACHIEVED HUMIDITY TARGETS RETAIN CONDITIONAL SAVINGS

Twelve converged eight-segment engineering-screen cases lie at or below the area gate. Selecting the lowest known total power for each achieved humidity ratio gives the following frontier.

Core outlet	Voltage [V]	Area [m^2]	Known total [kW]	Saving vs ERV+AC [kW / %]	Break-even [Wh/kg]
12.000 g/kg	0.10	6,519	490.475	53.265 / 9.80	25.379
11.000 g/kg	0.10	8,669	446.484	97.257 / 17.89	35.272
10.000 g/kg	0.50	7,066	429.256	114.485 / 21.06	43.779
9.000 g/kg	1.00	5,758	422.938	120.802 / 22.22	51.172
O30 (8.886 g/kg)	1.00	5,899	418.400	125.340 / 23.05	51.955

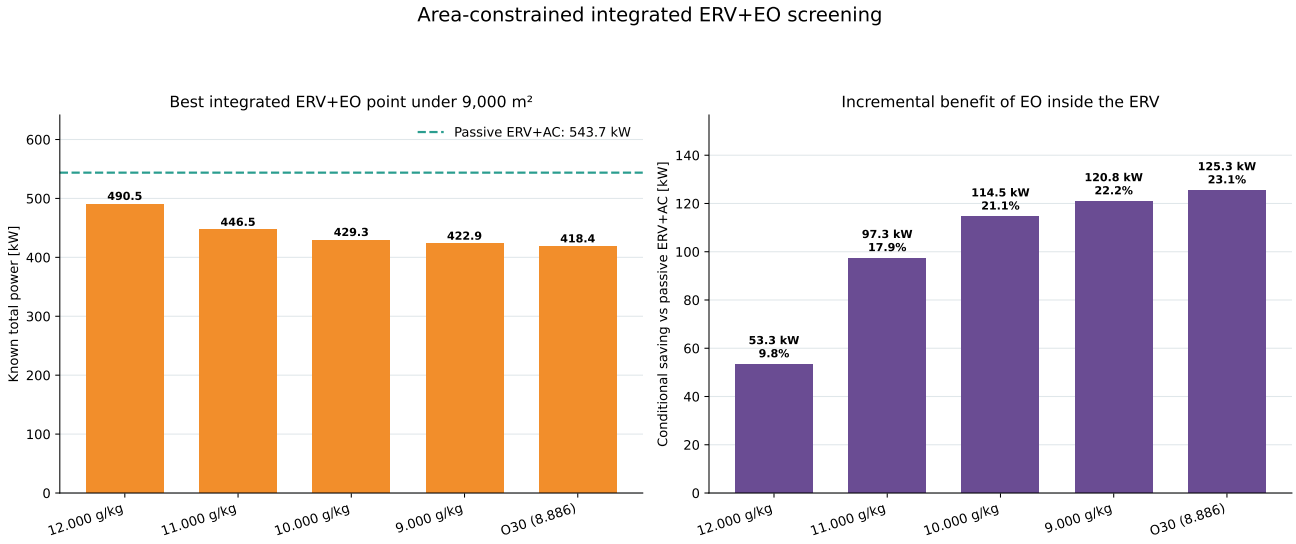


Figure 3: Best integrated ERV+EO point at each achieved humidity ratio below 9,000 m^2.

Deeper drying does not require monotonically increasing area because the selected voltage changes. Increasing voltage raises flux and can reduce area, but it also increases active-core power. The selected frontier is therefore a joint voltage–area–energy result, not an area-only trend.

The 8.000-g/kg target needs at least approximately 12,735 m^2 among the valid eight-segment cases, while the full supply-humidity target needs at least 17,253 m^2. Neither is accepted under the current gate.

3 Break-even and system-loss budget

THE OMITTED-LOAD ALLOWANCE IS FINITE AND ARCHITECTURE-SPECIFIC

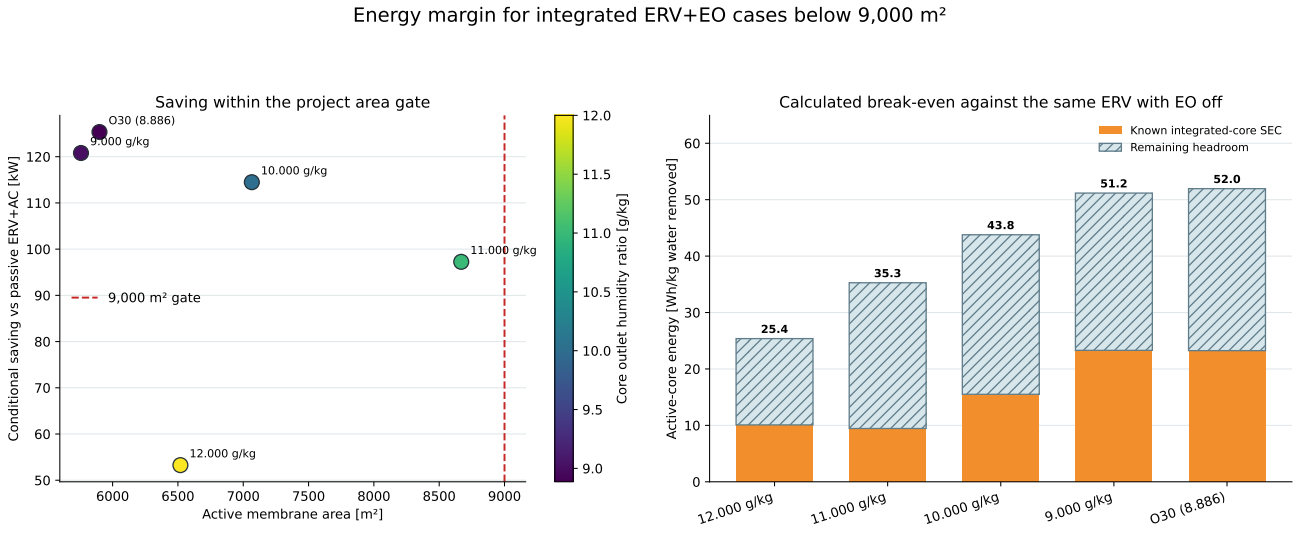


Figure 4: Conditional saving and calculated active-core break-even for the selected under-gate cases. Orange is known active-core SEC; hatching is the remaining allowance.

For O30, any combination of omitted receiver, heat-rejection, conversion, electrode, manifold and incremental fan loads above 125.340 kW removes the incremental advantage over passive ERV+AC. This power ceiling is equivalent to the 28.709-Wh/kg gap between known active-core SEC and calculated break-even. The limit must be treated as a design budget, not as a prediction that the missing subsystems will fit inside it.

No net-saving claim yet. Every positive value in this section is a known-power conditional window. The full-coupled-system gate is still open for all selected under-9,000-m^2 cases.

4 Model closure and membrane interpretation

THE REMAINING UNCERTAINTY IS EMPIRICAL AND SUBSYSTEM-SPECIFIC

The model phase has resolved several interpretation errors:

- a negative passive surface activity direction is not a thermodynamic rejection of electrically driven uphill EO transport;
- current density and membrane area are solved coordinates and must not be used as undocumented universal acceptance gates;
- water diffusivity alone does not select a membrane; the relevant quantity is a coupled bundle including sorption derivative, chemical water mobility/permeance, thickness, electro-osmotic drag, wet conductivity and compressed-stack resistance;
- a symmetric quasi-static $+J/0/-J$ replay produces no useful mean EO pumping beyond the passive state in the present constant-drag model; asymmetry, rectification or memory would require measured physics not yet in the model.

The direct active-core cases close the signed thermodynamic audit and the selected under-gate cases pass the current transport, robust-physics and engineering screens. They do not close receiver operation, conjugate thermal feedback, real manifolds, leakage, durability or product-scale current collection. A hypothetical joint material vector is therefore an experiment target, not a procurement specification.

Phase decision: GO for evidence generation. Stop unfocused parameter sweeps. Acquire same-sample material data and run a matched EO-off/EO-on integrated-ERV coupon experiment. Replay the model only after measured properties and subsystem power replace the open assumptions.

5 Vapour-cell experiment

CENTIMETRE-SCALE CONTROLLED FLOW, NOT A SHALLOW CLOSED STEADY CELL

The recommended first fixture uses a 50×50-mm active aperture (25 cm^2), an approximately 80×80-mm membrane coupon and 110×110-mm outer plates. Use interchangeable symmetric 1/2/4-mm gaps, with 2 mm nominal. At 1 m/s the nominal flow is 6.0 actual L/min per side. Use low-flow acquisition separately when humidity-sensor detectability requires it. Start with 8–10-mm insulating flow frames plus 12-mm metal backing/load plates; a load-bearing PEEK plate should start near 16 mm and be accepted only after loaded flatness and gap mapping.

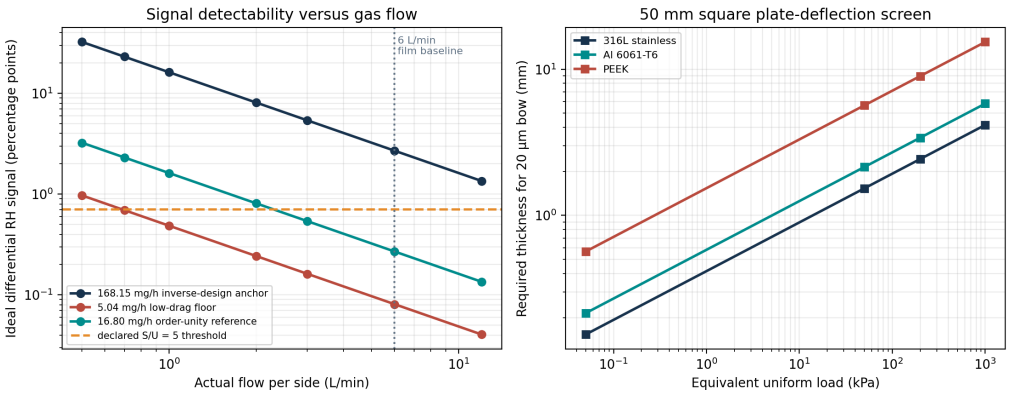


Figure 5: Coupon-cell dimension trade-offs supporting the 25-cm^2, 2-mm-gap starting fixture.

The main experimental controls are a split common inlet, ABBA polarity reversal, two-sided water balance, foil blanks, pressure control, gap/flow repeats, reference cross-switching and multiple aperture/perimeter inserts. Static temperature or

RH mismatch can dominate a one-direction estimate; polarity-odd analysis rejects the static component but not nonlinear or polarity-correlated drift. Closed chambers are transient inventory experiments and must not be described as fixed-boundary steady-state measurements.

6 Final decision and handoff

WHAT IS READY, WHAT IS NOT, AND WHAT HAPPENS NEXT

Decision	Status	Basis
Use passive ERV as primary control	GO	Matched 543.741-kW equal-service reference
Test EO integrated inside ERV to O30	GO	5,899 m^2; 125.340-kW conditional window
Use EO for O50 in this duty	NO DUTY	Passive ERV already passes O50
Claim full EO dehumidification below 9,000 m^2	NO-GO	Minimum valid screened full-duty area is about 17,253 m^2
Claim measured net energy saving	NOT YET	Receiver, heat rejection and hardware ledger remain open
Select a commercial membrane/module	NOT YET	Same-sample coupled properties and durability are unmeasured
Build controlled-flow coupon fixture	GO	Geometry, controls and error mechanisms are specified

The highest-value next actions are:

1. measure the same membrane's sorption, mobility/permeance, drag, wet conductivity, compressed ASR, thickness and mechanics;
2. run matched EO-off/EO-on ABBA tests at controlled film conditions;
3. close receiver, thermal, conversion and hydraulic power inside the 125.340-kW O30 allowance;
4. replay the integrated ERV model with measured data and decide whether the O30 window remains positive.

Final project conclusion. The project has identified a credible conditional opportunity and a concrete experiment capable of falsifying it. The correct claim is not that EO has already saved energy, but that EO integrated in the ERV can retain a 23.05% known-power window at O30 within the 9,000-m^2 gate, provided all omitted subsystem loads remain below the calculated allowance. This is ready for technical handoff and experimental continuation.

7 Controlled evidence basis

- `outputs/direct_active_eo_erv_study_v1/study/` — integrated active-core case register and audits;
- `outputs/integrated_erv_eo_opportunity_plots_v2/` — matched architecture, pairwise saving and full-duty screens;
- `outputs/integrated_erv_eo_under_9000_v1/` — selected under-gate cases and calculated break-even;
- `outputs/eo_vapor_cell_sizing_v1/` and `outputs/eo_vapor_boundary_analysis_v1/` — fixture sizing and boundary-condition analysis;
- `reports/weekly_findings_2026-08-24_to_2026-08-30/` — Revision 4 weekly evidence chronology;
- `reports/eo_vapor_boundary_analysis_v2_en/` — canonical English vapour-cell report.

End of controlled Version 9 report. All positive energy differences are conditional model windows and do not constitute material, module, product or commercial validation.